

1 Making Decisions under Uncertainty

2 Convergent Prior

Definition 2.1 (probabilistic model). We define a *probabilistic model* as a pair $(\mathbf{S}, \mathbf{P})$, where $\mathbf{S}$ is a sample space, and $\mathbf{P}$ is a set of *acceptable* probability measures on $\mathbf{S}$.

Definition 2.2 (ϵ -convergence). Let S and E be random variables in a probabilistic model $\mathcal{M}$. We say $S|E$ (S given E) ϵ -converges by a distribution P in $\mathcal{M}$, if for every distribution Q in $\mathcal{M}$ we have:

$$\mathbb{E}_Q [\mathbb{D}(P_{S|E} || Q_{S|E})] = \sum_{s \in \mathcal{S}} \sum_{e \in \mathcal{E}} \ln \frac{Q(s | e)}{P(s | e)} Q(s, e) < \epsilon. \quad (1)$$

Definition 2.3 (ϵ -convergent distribution). Let S be a random variable, and $\mathbf{E}$ be a set of random variables in a probabilistic model $\mathcal{M}$. We say P is an ϵ -convergent distribution for $S|\mathbf{E}$ in $\mathcal{M}$, if for all $E \in \mathbf{E}$, $S|E$ ϵ -converges by P in $\mathcal{M}$.

Definition 2.4 (convergence measure). Let S and E be random variables in a probabilistic model $\mathcal{M} = (\mathbf{S}, \mathbf{P})$. We define the *convergence measure* of $S|E$ as follows:

$$\mathbb{C}_{\mathcal{M}}[S|E] = \min_{P \in \mathbf{P}} \max_{Q \in \mathbf{P}} \sum_{s \in \mathcal{S}} \sum_{e \in \mathcal{E}} \ln \frac{Q(s | e)}{P(s | e)} Q(s, e). \quad (2)$$

Proposition. Let S be a random variable, and $\mathbf{E}$ be a set of random variables in a probabilistic model $\mathcal{M}$. If there exists $E \in \mathbf{E}$ such that $\mathbb{C}_{\mathcal{M}}[S|E] \geq \epsilon$, then there is no ϵ -convergent distribution for $S|\mathbf{E}$ in $\mathcal{M}$,

Theorem 1. Consider two probabilistic models $\mathcal{M}$ and $\mathcal{M}^*$. Let $\mathbf{E}$ be a countable set of random variables in $\mathcal{M}$, and E^* be a single random variable in $\mathcal{M}^*$. Suppose that there exists a random variable I in $\mathcal{M}^*$, such that for **every** probability measure P in $\mathcal{M}$, there exists a probability measure P^* in $\mathcal{M}^*$, such that for every $E_i \in \mathbf{E}$, every $s \in \mathcal{S}$ and every $e_i \in \mathcal{E}_i$, there exist some values s^*, e^* such that we have:

$$P(S = s, E_i = e_i) = P^*(S^* = s^*, E^* = e^* | I = i). \quad (3)$$

In addition, I and E^* are conditionally independent:

$$P^*(S^* | E^*) = P^*(S^* | E^*, I). \quad (4)$$

Then $S|\mathbf{E}$ ϵ -converges by P in $\mathcal{M}$, if $S^*|E^*$ ϵ -converges by P^* in $\mathcal{M}^*$.

Proof. Assume, by contradiction, that $S|\mathbf{E}$ does not ϵ -converge by P in $\mathcal{M}$. Therefore, there must exist Q and $E_i \in \mathbf{E}$ in $\mathcal{M}$ such that:

$$\mathbb{E}_Q [\mathbb{D}(P_{S|E_i} || Q_{S|E_i})] \geq \epsilon .$$

By using equations 3 and 4 we can rewrite the left hand side:

$$\begin{aligned} \mathbb{E}_Q [\mathbb{D}(P_{S|E_i} || Q_{S|E_i})] &= \sum_{s \in \mathcal{S}} \sum_{e_i \in \mathcal{E}_i} \ln \frac{Q(s | e_i)}{P(s | e_i)} Q(s, e_i) \\ &= \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*, I = i)}{P^*(s^* | e^*, I = i)} Q^*(s^*, e^* | I = i) \\ &= \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^* | I = i) . \end{aligned}$$

Hence, we have:

$$\sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^* | I = i) \geq \epsilon .$$

Since $Q^*(i)$ is a distribution and is non-negative for every i , we can multiply both sides of this inequality to $Q^*(i)$ and sum over all values of i :

$$\begin{aligned} \sum_i \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^*, I = i) &\geq \sum_i \epsilon Q^*(i) \\ \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} \sum_i Q^*(s^*, e^*, I = i) &\geq \epsilon \sum_i Q^*(i) \\ \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^*) &\geq \epsilon , \end{aligned}$$

which means $S^*|E^*$ does not ϵ -converge by P^* in $\mathcal{M}^*$. □

blahblahblah said Nobody [1]

References

- [1] Nobody Jr. My article, 2006.